

Demostración de tablas en MySQL con Docker

Se construye la imagen de Docker a partir del Dockerfile

Dockerfile realizado para la creación de la imagen.

```
Dockerfile > ...
1 FROM mysql:8.0
2
3 ENV MYSQL_ROOT_PASSWORD=root
4
5 # Declaración explícita de volumen
6 VOLUME ["/var/lib/mysql"]
7
8 COPY 01_create_database.sql /docker-entrypoint-initdb.d/
9 COPY 02_create_tables.sql /docker-entrypoint-initdb.d/
10 COPY 03_create_foreign_keys.sql /docker-entrypoint-initdb.d/
11 COPY 04_insert_data.sql /docker-entrypoint-initdb.d/
12
13 EXPOSE 3306
```

Construcción de la imagen con el uso del comando Docker build

```
arago@Alilap MINGW64 ~/Desktop/Docu/Modelo 5to/Base Datos/Parcial 1/Tarea 2
$ docker build -t pet-care .
[+] Building 0.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 388B
=> [internal] load metadata for docker.io/library/mysql:8.0
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/mysql:8.0@sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
=> => resolve docker.io/library/mysql:8.0@sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
=> [internal] load build context
=> => transferring context: 167B
=> CACHED [2/5] COPY 01_create_database.sql /docker-entrypoint-initdb.d/
=> CACHED [3/5] COPY 02_create_tables.sql /docker-entrypoint-initdb.d/
=> CACHED [4/5] COPY 03_create_foreign_keys.sql /docker-entrypoint-initdb.d/
=> CACHED [5/5] COPY 04_insert_data.sql /docker-entrypoint-initdb.d/
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:48ef21eb1effe6305ba654f8cd06c744f312af6a1be191852708064c0a6ac0d7
=> => exporting config sha256:eb2a68a33a0b01a8d479614f29556863fd7a81d8760584cfab8d73c4267beb7b
=> => exporting attestation sha256:374e0972a1db44dc8ee9907b50ecb854c0f21e575bd3c2060435af36e25f38a2
=> => exporting manifest list sha256:88259153ae9c50495479998544cca0522a2c66eb6f8f9bc5bbb5766c1e4fee98
=> => naming to docker.io/library/pet-care:latest
=> => unpacking to docker.io/library/pet-care:latest

1 warning found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "MYSQL_ROOT_PASSWORD") (line 3)

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/kegok5m31tgavtimckv63i728
```

Se levanta el contenedor de la imagen con el uso de Docker run.

```
arago@Alilap MINGW64 ~/Desktop/Docu/Modelo 5to/Base Datos/Parcial 1/Tarea 2
$ docker run -d --name pet-care -p 3306:3306 pet-care:latest
9c6fae14baff3a8ddea940ec0159d9e8687755fda781485a4af718b466554324

arago@Alilap MINGW64 ~/Desktop/Docu/Modelo 5to/Base Datos/Parcial 1/Tarea 2
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
9c6fae14baff   pet-care:latest   "docker-entrypoint.s..."   4 seconds ago   Up 4 seconds   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp   pet-care
```

Se prende el contenedor e ingresamos dentro del contenedor pet-care. Se loggea dentro de MySQL con mysql -u root -p.

```

arago@Alilap MINGW64 ~/Desktop/Docu/Modelo 5to/Base Datos/Parcial 1/Tarea 2
$ docker exec -it 9c6fae14baff bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>

```

Se muestran las bases de datos y se usa de “coronado-castaneda”

```

mysql> show databases;
+-----+
| Database |
+-----+
| coronado-castaneda |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> use coronado-castaneda;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> |

```

Se muestran las tablas dentro de la base de datos.

```

Database changed
mysql> show tables;
+-----+
| Tables_in_coronado-castaneda |
+-----+
| appointments |
| clients |
| drugs |
| pets |
| prescriptions |
| veterinarians |
+-----+
6 rows in set (0.00 sec)

mysql> |

```

Muestra de los registros dentro del contenedor de Docker

```

SELECT * FROM pets;
+----+-----+-----+-----+-----+-----+
| id | DNI      | name | lastName | phone | email |
+----+-----+-----+-----+-----+-----+
| 1 | 11111111-1 | Juan | Pérez   | 912345678 | juan.perez@email.com |
| 2 | 22222222-2 | María | Gómez   | 923456789 | maria.gomez@email.com |
| 3 | 33333333-3 | Carlos | López   | 934567890 | carlos.lopez@email.com |
| 4 | 44444444-4 | Ana | Martínez | 945678901 | ana.martinez@email.com |
| 5 | 55555555-5 | Pedro | Sánchez | 956789012 | pedro.sanchez@email.com |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM pets;
+----+-----+-----+-----+-----+-----+
| id | name | species | breed | birthday | client_id |
+----+-----+-----+-----+-----+-----+
| 1 | Firulaís | Perro | Labrador | 2020-05-10 | 1 |
| 2 | Michi | Gato | Siamés | 2021-08-15 | 2 |
| 3 | Rex | Perro | Pastor Alemán | 2019-03-22 | 1 |
| 4 | Pelusa | Conejo | Enano | 2022-01-05 | 3 |
| 5 | Thor | Perro | Bulldog | 2021-11-30 | 4 |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM veterinarians;
+----+-----+-----+-----+-----+-----+
| id | DNI      | name | lastName | specialty | phone |
+----+-----+-----+-----+-----+-----+
| 1 | 66666666-6 | Laura | Torres | Cirugía | 967890123 |
| 2 | 77777777-7 | Diego | Ramirez | Dermatología | 978901234 |
| 3 | 88888888-8 | Sofia | Castro | General | 989012345 |
| 4 | 99999999-9 | Javier | Morales | Oftalmología | 990123456 |
| 5 | 10101010-0 | Elena | Rojas | Cardiología | 901234567 |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM appointments;
+----+-----+-----+-----+-----+-----+-----+
| id | appointment_date | diagnostic | cost | veterinarian_id | pet_id |
+----+-----+-----+-----+-----+-----+-----+
| 1 | 2026-03-01 10:00:00 | Chequeo general y vacunación | 35.00 | 3 | 1 |
| 2 | 2026-03-01 11:30:00 | Infección cutánea en la oreja | 45.00 | 2 | 2 |
| 3 | 2026-03-02 09:15:00 | Revisión por cojera en pata posterior | 50.00 | 1 | 3 |
| 4 | 2026-03-02 14:00:00 | Corte de uñas y revisión dental | 25.00 | 3 | 4 |
| 5 | 2026-03-03 16:45:00 | Problema respiratorio leve | 40.00 | 5 | 5 |
+----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM drugs;
SELECT * FROM prescriptions;
+----+-----+-----+-----+-----+-----+
| id | code | name | laboratory | unit_price |
+----+-----+-----+-----+-----+-----+
| 1 | DRUG001 | Amoxicilina 250mg | Laboratorio Alfa | 12.50 |
| 2 | DRUG002 | Meloxicam Jarabe | VetPharm | 18.00 |
| 3 | DRUG003 | Ketoconazol Crema | DermoVet | 15.20 |
| 4 | DRUG004 | Antiparasitario Interno | PetHealth | 8.50 |
| 5 | DRUG005 | Gotas Oftálmicas | OftaVet | 22.00 |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

```

mysql> SELECT * FROM prescriptions;
+----+-----+-----+-----+-----+
| id | drug_id | appointment_id | quantity | instructions |
+----+-----+-----+-----+-----+
| 1 | 1 | 1 | 10 | 1 pastilla cada 12 horas por 5 días |
| 2 | 3 | 2 | 1 | Aplicar en la zona afectada 2 veces al día por 7 días |
| 3 | 2 | 3 | 1 | Dar 2ml cada 24 horas por 3 días |
| 4 | 4 | 4 | 1 | Administrar una dosis única por vía oral |
| 5 | 1 | 5 | 14 | 1 pastilla cada 12 horas por 7 días |
+----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

```

-- INNER JOIN veterinarios y ON el veterinario = v.id;
+----+-----+-----+-----+-----+-----+-----+
| ID_Cita | Fecha_Hora | Mascota | Veterinario | Especialidad | Diagnostico | Costo |
+----+-----+-----+-----+-----+-----+-----+
| 1 | 2026-03-01 10:00:00 | Firulaís | Sofia Castro | General | Chequeo general y vacunación | 35.00 |
| 2 | 2026-03-01 11:30:00 | Michi | Diego Ramirez | Dermatología | Infección cutánea en la oreja | 45.00 |
| 3 | 2026-03-02 09:15:00 | Rex | Laura Torres | Cirugía | Revisión por cojera en pata posterior | 50.00 |
| 4 | 2026-03-02 14:00:00 | Pelusa | Sofia Castro | General | Corte de uñas y revisión dental | 25.00 |
| 5 | 2026-03-03 16:45:00 | Thor | Elena Rojas | Cardiología | Problema respiratorio leve | 40.00 |
+----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```